PROGRAMMING FOR PROBLEM SOLVING

I B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6CS02	ESC	3	-	-	3	40	60	100
Contact Classes: 64	Tutorial Classes: Nil	Practical Classes: Nil				Total Classes: 64		

COURSE OBJECTIVES

The course will enable the students to:

1. To familiarize with the syntax and semantics of C programming language.
2. To learn the usage of structured programming approach in solving problems.
3. To use arrays, pointers, strings and structures in solving problems.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Apply algorithmic thinking to understand, define and solve problems
2. Develop computer programs using programming constructs and control structures and to use arrays to develop C programs
3. Decompose a problem into functions to develop modular reusable code and to use pointers to solve complex problems.
4. Use Strings and structures to formulate algorithms and programs.
5. Use FILE to perform read and write operations.

UNIT - I	INTRODUCTION - PROBLEM SOLVING AND ALGORITHMIC THINKING& INTRODUCTION TO C LANGUAGE	CLASSES: 12

Algorithm -Definition, Characteristics of Algorithm. Constituents of algorithms**: -** Sequence, Selection and Repetition. Algorithm with Example: Roots Of a Quadratic Equations, Minimum and Maximum Numbers of a Given Set, Given number is prime number or not, given integer is palindrome or not, etc. Flowchart/Pseudo Code with examples.
Introduction To C Language: Structure of C Program, Data Types, Input and Output statements, Operators, Precedence and Associativity of operators, Evaluation of Expressions, Type Conversions In Expressions.

UNIT - II	CONTROL STRUCTURES AND ARRAYS	CLASSES: 15

Control structures: Decision statements; if and switch statement; **Loop control statements:** while, for and do while loops, Jump statements: break, continue, goto statements.
Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi-dimensional arrays

UNIT - III	FUNCTIONS AND POINTERS	CLASSES: 17

Functions: Function definition, Types of Functions: User defined and built-in Functions, Advantages of User Defined Functions. Parameter passing in functions**:** Call by value, Call by reference, Passing arrays to functions, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Towers of Hanoi etc... Storage classes.
Pointers: Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, Functions returning pointers, Dynamic memory allocation.

UNIT - IV	STRINGS AND USER DEFINED DATA TYPES	CLASSES: 10

Strings: Array of characters, variable length character strings, inputting character strings, character library functions, String Handling Functions, Arrays Of Strings
Structures and Unions: Structure definition, initialization, accessing structures, nested structures, arrays of structures, Structures and functions, Self-referential structures, unions, typedef, enumerations.

UNIT - V	FILE HANDLING,SEARCHING AND SORTING	CLASSES: 10

File Handling: Command Line Arguments, File Modes, And Basic File Operations Read, Write and Append, Example Programs. Random Access Using fseek, ftell and rewind Functions.
Basic Searching And Sorting Algorithms: Linear and Binary Search, Bubble Sort, Insertion Sort, Quick Sort.

TEXT BOOKS

1. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition)
2. Byron Gottfried, “Programming with C”, Schaum's Outlines Series, McGraw Hill Education, 3rdedition, 2017.
3. Programming in C E. Balagurusamy Edition 3 Publisher Tata McGraw-Hill Publishing, 1990

REFERENCE BOOKS

1. W. Kernighan Brian, Dennis M. Ritchie, “The C Programming Language”, PHI Learning, 2nd Edition, 1988.
2. YashavantKanetkar, “Exploring C”, BPB Publishers, 2nd Edition, 2003.
3. Schildt Herbert, “C: The Complete Reference”, Tata McGraw Hill Education, 4th Edition, 2014.4. R. S. Bichkar, “Programming with C”, Universities Press, 2nd Edition, 2012.
4. Dey Pradeep, Manas Ghosh, “Computer Fundamentals and Programming in C”, Oxford University Press, 2nd Edition, 2006.
5. Stephen G. Kochan, “Programming in C”, Addison-Wesley Professional, 4th Edition, 2014.

WEB REFERENCES

1. https://en.wikipedia.org/wiki/Computational_thinking
2. https://nptel.ac.in/courses/106/104/106104128/
3. https://en.cppreference.com/w/c/language
4. https://www.learn-c.org/

E-TEXT BOOKS

1. https://slidelegend.com/queue/computational-thinking-for-the-modern-problem solver_59d6f01e1723ddb0c7a0df47.html
2. http://flowgorithm.altervista.org/#elf_l1_Lw
3. http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm

MOOC COURSE

1. https://www.coursera.org/learn/computational-thinking-problem-solving
2. https://onlinecourses.nptel.ac.in/noc18_cs33/preview
3. https://www.alison.com/courses/Introduction-to-Programming-in-c
4. http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective programming-in-c-and-c-january-iap-2014/index.html